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Probability and Stochastic Processes

CAT 4

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1. Consider $M/M/1$: This is the simplest queueing system, where the arrival process is Poisson and the service mechanism is Exponential. The system consists of one server, an unlimited FIFO queue, and an unlimited customer population

Derive the formulas for:

- (a) Probability that there are n entities in the system.
- (b) Average entities in the system (L_s).
- (c) Average entities in the queue (L_q).
- (d) Average waiting time in the system (W_s).
- (e) Average waiting time in the queue (W_q).

Assumptions

- **Arrival process:** Poisson with rate $\lambda > 0$.
- **Service process:** i.i.d. exponential service times with rate $\mu > 0$.
- **Single server, FIFO discipline, infinite buffer, infinite population.**
- **State:** $N(t) \in \{0, 1, 2, \dots\}$ is the number of entities (customers) in the system (waiting + in service) at time t .
- **Stability (ergodicity):** $\rho\lambda/\mu < 1$ so that a steady state exists.
- This is a *birth-death process* with birth rate λ and death rate μ whenever the system is non empty.

Steady-State Balance and Normalization

Let $\pi_n \lim_{t \rightarrow \infty} \mathbb{P}\{N(t) = n\}$ be the steady-state probability of n entities in the system.

Step 1: Local balance (birth-death recursion)

For $n \geq 0$, the *detailed balance* (or local balance) equations are

$$\lambda \pi_n = \mu \pi_{n+1} \quad \implies \quad \pi_{n+1} = \frac{\lambda}{\mu} \pi_n = \rho \pi_n.$$

By induction,

$$\pi_n = \rho^n \pi_0, \quad n = 0, 1, 2, \dots$$

Step 2: Normalization

Using $\sum_{n=0}^{\infty} \pi_n = 1$,

$$1 = \sum_{n=0}^{\infty} \rho^n \pi_0 = \pi_0 \sum_{n=0}^{\infty} \rho^n = \pi_0 \frac{1}{1-\rho} \quad (\text{since } \rho < 1),$$

hence

$$\pi_0 = 1 - \rho.$$

Therefore,

$$\boxed{\pi_n = (1 - \rho) \rho^n, \quad n = 0, 1, 2, \dots}.$$

(a) Probability of n entities in the system

From the derivation above,

$$\mathbb{P}\{N = n\} = \pi_n = (1 - \rho) \rho^n, \quad \rho = \frac{\lambda}{\mu} < 1.$$

(b) Average number in the system L_s

Compute

$$L_s \mathbb{E}[N] = \sum_{n=0}^{\infty} n \pi_n = (1 - \rho) \sum_{n=0}^{\infty} n \rho^n.$$

Recall the power series identity for $|x| < 1$:

$$\sum_{n=0}^{\infty} n x^n = \frac{x}{(1 - x)^2}.$$

Thus,

$$L_s = (1 - \rho) \cdot \frac{\rho}{(1 - \rho)^2} = \frac{\rho}{1 - \rho}.$$

Hence

$$L_s = \frac{\rho}{1 - \rho} = \frac{\lambda}{\mu - \lambda}.$$

(c) Average number in the queue L_q

Decompose $N = N_q + N_s$, where N_q is the number *waiting* and $N_s \in \{0, 1\}$ is the number *in service*. By steady state, the server is busy with probability ρ ; thus $\mathbb{E}[N_s] = \rho$. Therefore,

$$L_q \mathbb{E}[N_q] = \mathbb{E}[N] - \mathbb{E}[N_s] = L_s - \rho = \frac{\rho}{1 - \rho} - \rho = \frac{\rho^2}{1 - \rho}.$$

Equivalently, in rates,

$$L_q = \frac{\rho^2}{1 - \rho} = \frac{\lambda^2}{\mu(\mu - \lambda)}.$$

(d) Average waiting time in the system W_s

By Little's Law applied to the *entire system*:

$$L_s = \lambda W_s \implies W_s = \frac{L_s}{\lambda} = \frac{\rho}{(1 - \rho)\lambda}.$$

Using $\rho = \lambda/\mu$,

$$W_s = \frac{\lambda/\mu}{(1 - \lambda/\mu)\lambda} = \frac{1}{\mu - \lambda}.$$

Hence

$$W_s = \frac{1}{\mu - \lambda}.$$

(e) Average waiting time in the queue W_q

By Little's Law applied to the *queue*:

$$L_q = \lambda W_q \implies W_q = \frac{L_q}{\lambda} = \frac{1}{\lambda} \cdot \frac{\rho^2}{1 - \rho}.$$

Using $\rho = \lambda/\mu$,

$$W_q = \frac{1}{\lambda} \cdot \frac{(\lambda/\mu)^2}{1 - \lambda/\mu} = \frac{\lambda}{\mu^2} \cdot \frac{1}{1 - \lambda/\mu} = \frac{\lambda}{\mu^2} \cdot \frac{\mu}{\mu - \lambda} = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{\rho}{\mu - \lambda}.$$

Thus

$$W_q = \frac{\rho}{\mu - \lambda} = \frac{\lambda}{\mu(\mu - \lambda)}.$$

2. Calls arrive at a call booth at the rate 110 calls per second. The average time to respond to a call is 200 seconds (i.e., mean service time = 200 s).

- (a) Determine whether the system is stable (does a steady-state exist?) and show the computation.
- (b) If the system were stable, compute the steady-state probability that at least 5 calls are present in the system (waiting + in service). If it is unstable, explain what that implies for that probability.

Given

- Arrival rate: $\lambda = 110$ calls/second.
- Mean service time: $\mathbb{E}[S] = 200$ seconds $\Rightarrow \mu = \frac{1}{\mathbb{E}[S]} = \frac{1}{200}$ per second = 0.005 s^{-1} .

(a) Stability (existence of steady state)

For an $M/M/1$ queue, the utilization is

$$\rho = \frac{\lambda}{\mu}.$$

Compute:

$$\rho = \frac{110}{0.005} = 110 \times \frac{1}{0.005} = 110 \times 200 = 22,000.$$

Stability requires $\rho < 1$. Here $\rho = 22,000 \gg 1$, so **the system is *unstable*** and a steady state does *not* exist.

(b) Probability of at least 5 calls in steady state

In a stable $M/M/1$ with $\rho < 1$, the steady-state tail is

$$\mathbb{P}\{N \geq k\} = \rho^k.$$

Hence, if (hypothetically) the system were stable,

$$\mathbb{P}\{N \geq 5\} = \rho^5.$$

However, with the given parameters the system is **not** stable, so no steady-state distribution exists and the probability $\mathbb{P}\{N \geq 5\}$ in steady state is *undefined*. Operationally, as time $t \rightarrow \infty$, the number in system grows without bound, so the *transient* probability $\mathbb{P}\{N(t) \geq 5\}$ approaches 1.

Conclusion

- (a) Unstable because $\rho = 22,000 > 1$.
- (b) No steady-state probability exists; the long-run probability of having at least 5 calls tends to 1.

3. Customers arrive at a help desk according to a Poisson process at 4 customers per minute. Each customer's service time is exponential with mean 1/6 minute (i.e., service rate $\mu = 6$ per minute).

- (a) Compute the traffic intensity ρ .
- (b) Find the average waiting time in queue W_q and the average time in system W .
- (c) What is the probability that an arriving customer must wait?
- (d) What is the probability that a customer's waiting time in queue exceeds 1 minute? (Give the exact expression and numeric value.)

Given

$$\lambda = 4 \text{ customers/min}, \quad \mu = 6 \text{ customers/min} \quad (\text{mean service time} = 1/\mu = 1/6 \text{ min}).$$

For an $M/M/1$ queue, define $\rho = \lambda/\mu$ and require $\rho < 1$ for stability.

(a) Traffic intensity ρ

$$\rho = \frac{\lambda}{\mu} = \frac{4}{6} = \frac{2}{3}.$$

(b) Average waiting time in queue W_q and in system W

Key $M/M/1$ formulas:

$$W = \frac{1}{\mu - \lambda}, \quad W_q = \frac{\rho}{\mu - \lambda} = \frac{\lambda}{\mu(\mu - \lambda)}.$$

Compute $\mu - \lambda = 6 - 4 = 2$:

$$W = \frac{1}{2} \text{ min} = 0.5 \text{ min} = 30 \text{ s},$$

$$W_q = \frac{\rho}{\mu - \lambda} = \frac{\frac{2}{3}}{2} = \frac{1}{3} \text{ min} \approx 0.333 \text{ min} = 20 \text{ s}.$$

(Consistency check via Little's Law: $L = \lambda W = 4 \times 0.5 = 2$ and $L_q = \lambda W_q = 4 \times \frac{1}{3} = \frac{4}{3}$.)

(c) Probability that an arriving customer must wait

By PASTA (Poisson Arrivals See Time Averages), the probability the server is busy upon arrival equals the utilization:

$$\mathbb{P}\{\text{must wait}\} = \rho = \frac{2}{3}.$$

(d) Tail of the waiting-time-in-queue distribution

In $M/M/1$, the (unconditional) waiting time in queue W_q has a point mass at 0 of size $1 - \rho$ and, conditional on being positive, is exponential with rate $\mu - \lambda$. For $t > 0$,

$$\mathbb{P}\{W_q > t\} = \rho e^{-(\mu - \lambda)t}.$$

At $t = 1$ minute and $\mu - \lambda = 2$:

$$\mathbb{P}\{W_q > 1\} = \rho e^{-2} = \frac{2}{3} e^{-2}.$$

Numerically, using $e^{-2} \approx 0.135335283$,

$$\mathbb{P}\{W_q > 1\} \approx \frac{2}{3} \times 0.135335283 \approx 0.09022.$$

4. Jobs arrive at a server at rate $\lambda = 2$ jobs/minute. Service times are exponential with mean 0.15 minute (so $\mu = 6.6667$ per minute).

- (a) Compute the steady-state probability $P\{N = n\}$ for an arbitrary n .
- (b) Using the expression in (a), compute the probability that there are at least 3 jobs in the system: $P\{N \geq 3\}$.
- (c) Compute the long-run fraction of time the server is idle.

Given

$$\lambda = 2 \text{ jobs/min,} \quad \text{mean service time} = 0.15 \text{ min} \Rightarrow \mu = \frac{1}{0.15} = \frac{20}{3} \approx 6.6667 \text{ jobs/min.}$$

Utilization (traffic intensity):

$$\rho = \frac{\lambda}{\mu} = \frac{2}{20/3} = \frac{2 \cdot 3}{20} = \frac{3}{10} = 0.3.$$

Since $\rho < 1$, the system is stable and steady-state results apply.

(a) Steady-state probability $P\{N = n\}$

For an $M/M/1$ queue,

$$P\{N = n\} = (1 - \rho)\rho^n, \quad n = 0, 1, 2, \dots$$

Hence with $\rho = \frac{3}{10}$,

$$P\{N = n\} = \left(1 - \frac{3}{10}\right) \left(\frac{3}{10}\right)^n = \frac{7}{10} \left(\frac{3}{10}\right)^n.$$

(b) Probability of at least 3 jobs: $P\{N \geq 3\}$

Use the geometric tail for $M/M/1$:

$$P\{N \geq 3\} = \sum_{n=3}^{\infty} (1 - \rho)\rho^n = (1 - \rho) \frac{\rho^3}{1 - \rho} = \rho^3.$$

Therefore,

$$P\{N \geq 3\} = \rho^3 = \left(\frac{3}{10}\right)^3 = \frac{27}{1000} = 0.027.$$

(c) Fraction of time the server is idle

The server is idle exactly when the system is empty, so

$$P\{\text{idle}\} = P\{N = 0\} = 1 - \rho = \left[1 - \frac{3}{10} = \frac{7}{10} = 0.7\right].$$

5. A web server is modeled as an $M/M/1$ queue. The system should have, on average, no more than 2 requests in the system (waiting + service) to meet SLAs. The service rate is $\mu = 20$ requests per minute.

- (a) Find the maximum arrival rate $\lambda_{\max}$ (in requests per minute) that satisfies $L \leq 2$.
- (b) If the arrival rate is set to $\lambda = \frac{1}{2}\lambda_{\max}$, compute the expected waiting time in queue W_q and the probability that an arrival finds the server busy.

M/M/1 Capacity Planning for SLA: $L \leq 2$, $\mu = 20/\text{min}$

Background formulas (M/M/1, $\rho = \lambda/\mu < 1$)

$$L = \frac{\rho}{1-\rho} = \frac{\lambda}{\mu-\lambda}, \quad W = \frac{1}{\mu-\lambda}, \quad W_q = \frac{\rho}{\mu-\lambda} = \frac{\lambda}{\mu(\mu-\lambda)}, \quad \mathbb{P}\{\text{busy}\} = \rho.$$

(a) Maximum arrival rate $\lambda_{\max}$ subject to $L \leq 2$

Impose the SLA on the mean number in system:

$$L = \frac{\lambda}{\mu-\lambda} \leq 2 \implies \lambda \leq 2(\mu-\lambda) \implies 3\lambda \leq 2\mu.$$

With $\mu = 20/\text{min}$,

$$3\lambda \leq 40 \implies \boxed{\lambda_{\max} = \frac{40}{3} \text{ requests/min} \approx 13.333 \text{ /min}}.$$

(Checks: $\lambda_{\max} < \mu$ so stability holds.)

(b) Set $\lambda = \frac{1}{2}\lambda_{\max}$ and compute W_q and $\mathbb{P}\{\text{busy}\}$

Half of the maximum rate:

$$\lambda = \frac{1}{2}\lambda_{\max} = \frac{1}{2} \cdot \frac{40}{3} = \frac{20}{3} \text{ /min.}$$

Traffic intensity:

$$\rho = \frac{\lambda}{\mu} = \frac{\frac{20}{3}}{20} = \frac{1}{3}.$$

Gap to capacity:

$$\mu - \lambda = 20 - \frac{20}{3} = \frac{40}{3} \text{ /min.}$$

Average waiting time in queue:

$$W_q = \frac{\rho}{\mu-\lambda} = \frac{\frac{1}{3}}{\frac{40}{3}} = \frac{1}{40} \text{ min} = 0.025 \text{ min} = \boxed{1.5 \text{ seconds}}.$$

(For reference, $W = 1/(\mu-\lambda) = \frac{3}{40} \text{ min} = 0.075 \text{ min} = 4.5 \text{ s}$, and $L = \lambda W = \frac{1}{2}$ as expected.)

Probability an arrival finds the server busy (by PASTA):

$$\boxed{\mathbb{P}\{\text{busy}\} = \rho = \frac{1}{3} \approx 0.333}.$$

Answers

$$\boxed{\lambda_{\max} = \frac{40}{3} \text{ /min} (\approx 13.333)}, \quad \boxed{W_q = \frac{1}{40} \text{ min} = 1.5 \text{ s}}, \quad \boxed{\mathbb{P}\{\text{busy}\} = \frac{1}{3}}.$$